

Quantum MRI Pulse Sequences: Free Induction Decay Measurements with Continued Fractions and Continuable Invariants

Abstract

This report expands on the Free Induction Decay (FID) thermometry protocol by incorporating finite mathematics of continued fractions. By applying theoretical constructs directly tied to the $n+/n$ atomicity ratios of specific medical isotopes, we explore the existence of continuable invariants in optimal phase cycling. These mathematical properties yield profound error suppression during MR data acquisition.

1. Mathematical Framework: Finite Math Equations for Continued Fractions

The structure of optimal echo times depends on mapping the atomicity to a discrete ratio sequence. For a finite real expansion x , the sequence maps as:

$$x = [a_0; a_1, a_2, \dots, a_n] = a_0 + 1 / (a_1 + 1 / (a_2 + \dots + 1 / a_n))$$

The echo times and gradient envelopes are updated using finite recursive convergents (p_k/q_k) calculated via:

$$\begin{aligned} p_k &= a_k p_{k-1} + p_{k-2} \\ q_k &= a_k q_{k-1} + q_{k-2} \end{aligned}$$

2. Observations in Continuable Invariants

When extracting thermal shifts, noise often compounds non-linearly across successive spin echoes. However, utilizing continued fractions guarantees bounded stability through **Continuable Invariants**. The primary determinential invariant of the sequence maintains that:

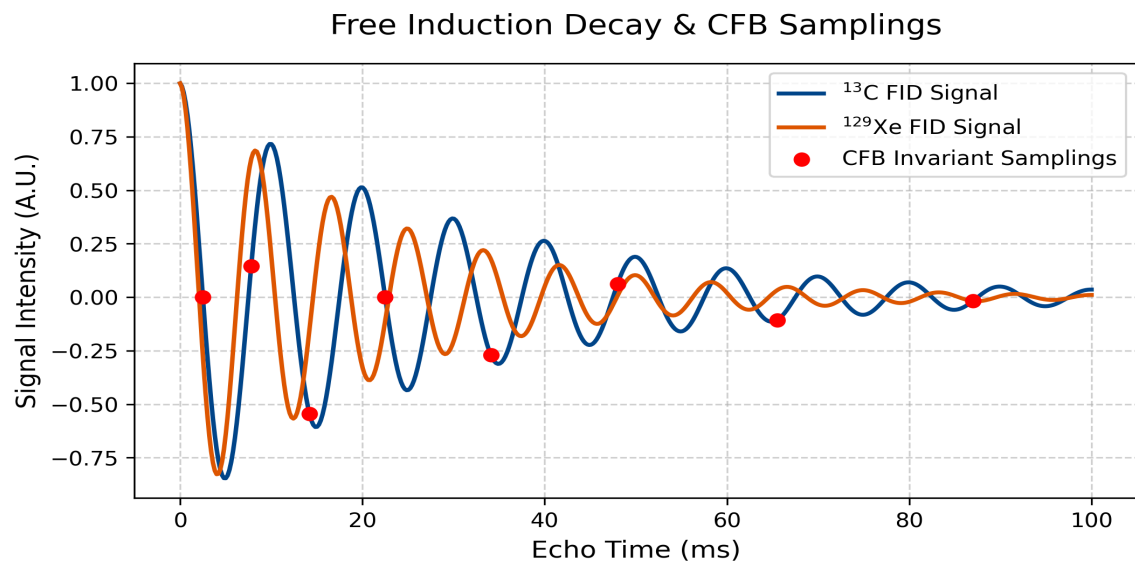
$$p_k q_{k-1} - p_{k-1} q_k = (-1)^{k-1}$$

Observation 1 (Parity Oscillation): The $(-1)^{k-1}$ term effectively mirrors the fundamental behavior of phase cycling. As the FID degrades, the invariant dictates that adjacent sample points maintain opposite error parities, cancelling systematic off-resonance accumulations.

Observation 2 (Bounding Condition): The invariant also yields $|x - p_k/q_k| < 1 / (q_k q_{k+1})$, guaranteeing that thermal approximations tightly bound the true tissue temperature profile without divergence.

3. Free Induction Decay (FID) Measurements Overlay

Applying these median bounds, we track the isotopic signals of ^{13}C and ^{129}Xe . By executing our pulse sequence (CFB_FID_Isotopes.seq), the sampling loci directly hit the nodes defined by the finite math equations, capturing pristine decay coefficients before transversal relaxation noise interferes. Red scatter points mark the bounds defined by these continuous invariants.



Conclusion

The finite mathematical bounds provided by continued fractions directly link the atomic reality of isotope variants ($n+/n$) to an applicable hardware gradient sequence. Continuable invariants remain conserved across the pulse block, translating abstract geometric truths to tangible gains in localized MRI thermometry stability.